

EULER'S THEOREM

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Euler's Theorem: For $n \in \mathbb{N}$ and $a \in \mathbb{Z}$ with $\gcd(n, a) = 1$, we have
 $a^{\varphi(n)} \equiv 1 \pmod{n}$.

proof: Consider $S = \{x \in \mathbb{N} \mid 1 \leq x \leq n, \gcd(x, n) = 1\}$
 $= \{x_1, x_2, \dots, x_{\varphi(n)}\}$

Take $aS = \{ax_1, ax_2, \dots, ax_{\varphi(n)}\}$.

claim: $\gcd(ax_i, n) = 1$. Otherwise there would be
① a prime p such that $p \mid ax_i$ and $p \mid n$. number
factor-
ization
 $\Rightarrow p \mid a$ or $p \mid x_i$, contradicts $\gcd(a, n) = 1 = \gcd(x_i, n)$.

claim: No 2 elements of aS are congruent mod n .

② Suppose $ax_i \equiv ax_j \pmod{n}$
 $a(x_i - x_j) \equiv 0 \pmod{n} \Rightarrow n \mid a(x_i - x_j)$
 $\Rightarrow n \mid x_i - x_j$. But $x_i - x_j < n \Rightarrow x_i - x_j = 0$
 $\Rightarrow x_i = x_j$

③+④
 $\Rightarrow aS \equiv S \pmod{n}$

This means that S and aS contain the same residues mod n .

$$\Rightarrow ax_1 \cdot ax_2 \cdot \dots \cdot ax_{\varphi(n)} \equiv x_1 \cdot \dots \cdot x_{\varphi(n)} \pmod{n}$$

Set $X := x_1 \cdot \dots \cdot x_n$; Note $\gcd(n, X) = 1 \Rightarrow \exists X^{-1} \pmod{n}$

$$\Rightarrow a^{\varphi(n)} X \equiv X \pmod{n}, \text{ multiply by } X^{-1}$$

$$\Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n} \quad \square$$